ARTIFICIAL INTELLIGENCE

UNIT-V

EXPERT SYSTEMS

Syllabus:

Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells.

Learning Outcomes:

At the end of the unit, students will be able to

- made to introduce the concepts of Expert Systems.

Module 1: Architecture of expert systems

Module 2: Typical expert systems

1. EXPERT SYSTEM

- The development of special-purpose computer programs, systems that were expert in some narrow problem area. These programs were called expert systems.

1.1 Architecture of expert systems

- At first, designing and building an expert system was considered as creative effort than a scientific enterprise.
- The process of building an expert system is often called knowledge engineering. It typically involves a special form of interaction between the expert-system builder, called the knowledge engineer, and one or more human experts in some problem area.
- The knowledge engineer "extracts" from the human experts their procedures, strategies, and rules of thumb for problem solving, and builds this knowledge into the expert system, as shown in Figure 1.

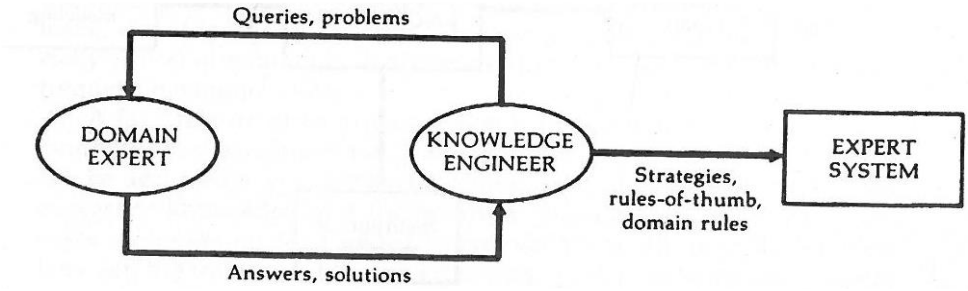


Figure 1: Knowledge engineering: transferring knowledge from an expert to a computer program

- The result is a computer program that solves problems in much the same manner as the human experts.
- An expert is a person who, because of training and experience, is able to do things the rest of us cannot.
- Experts are not only proficient but also smooth and efficient in the actions they take.
- Knowledge engineering relies heavily on the study of human experts in order to develop intelligent, skilled programs.

1.2 Roles of expert systems

- Expert systems solve problems that are normally solved by human "experts."
- To solve expert-level problems, expert systems need access to a substantial domain knowledge base, which must be built as efficiently as possible.
- Expert systems also need to exploit one or more reasoning mechanisms to apply their knowledge to the problems they are given. Then they need a mechanism for explaining what they have done to the users who depend on them.
- One way to look at expert systems is that they represent applied AI in a very broad sense.
- The problems that expert systems deal with are highly diverse. There are some general issues that arise across these varying domains. But it also turns out that there are powerful techniques that can be defined for specific classes of problems.
- The main players in the expert system game are the expert system, the domain expert, the knowledge engineer, the expert-system-building tool, and the user. Their basic roles and their relationship to each other are summarized in Figure 2.

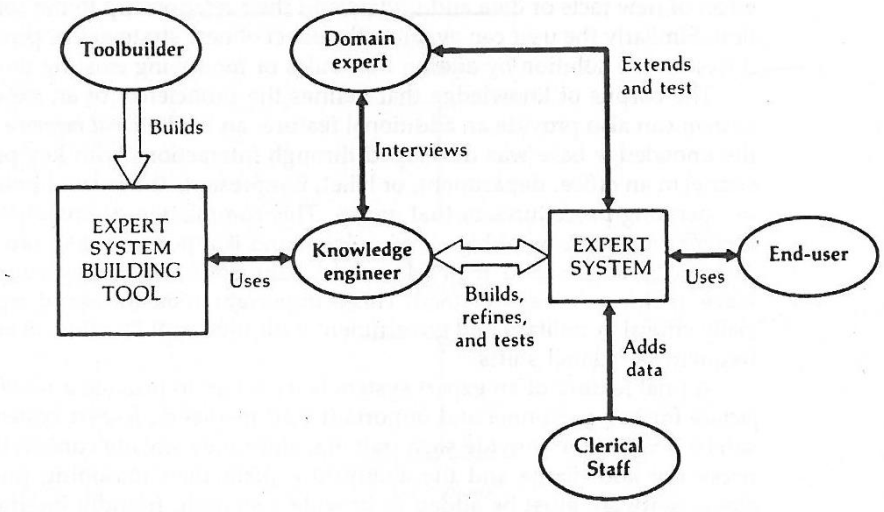


Figure 2: The players in the expert system game

1.2.1 Expert system

- The expert system is the collection of programs or computer software that solves problems in the domain of interest.
- It's called a system rather than just a program because it contains both a problem-solving component and a support component.

1.2.2 Domain expert

- The domain or area expert is a knowledgeable person with a reputation for producing good solutions to problems in a particular field.
- The expert uses tricks and shortcuts to make the search for a solution more efficient, and the expert system models these problem-solving strategies.

1.2.3 Knowledge engineer

- The knowledge engineer is a human, usually with a background in computer science and AI, who knows how to build expert systems.
- The knowledge engineer interviews the experts, organizes the knowledge, decides how it should be represented in the expert system, and may help programmers to write the code.

1.2.4 Expert-system-building tool

- The expert-system-building tool is the programming language used by the knowledge engineer or programmer to build the expert system.
- In AI terminology, the term tool usually refers both to the programming language and to the support environment used to build the expert system.

1.2.5 User

- The user is the human who uses the expert system once it is developed.

3. KNOWLEDGE ACQUISITION META KNOWLEDGE HEURISTICS

- After the initial system is built, it must be iteratively refined until it approximates expert-level performance. This process is expensive and time-consuming, so it is worthwhile to look for more automatic ways of constructing expert knowledge bases.
- While no totally automatic knowledge acquisition systems yet exist, there are many programs that interact with domain experts to extract expert knowledge efficiently.
- These programs provide support for the following activities:
    1. Entering knowledge
    2. Maintaining knowledge base consistency
    3. Ensuring knowledge base completeness

- The most useful knowledge acquisition programs are those that are restricted to a particular problem-solving paradigm, e.g., diagnosis or design.
- It is important to be able to enumerate the roles that knowledge can play in the problem-solving process.
- For example, if the paradigm is diagnosis, then the program can structure its knowledge base around symptoms, hypotheses, and causes. It can identify symptoms for which the expert has not yet provided causes.
- Since one symptom may have multiple causes, the program can ask for knowledge about how to decide when one hypothesis is better than another.
- MOLE is a knowledge acquisition system for heuristic classification problems, such as diagnosing diseases. In particular, it is used in conjunction with the cover-and-differentiate problem-solving method.
- An expert system produced by MOLE accepts input data, comes up with a set of candidate explanations or classifications that cover (or explain) the data, then uses differentiating knowledge to determine which one is best.
- MOLE interacts with a domain expert to produce a knowledge base that a system called MOLE-p (for MOLE-performance) uses to solve problems.
- The acquisition proceeds through several steps:

3.1 Initial knowledge base construction

- MOLE asks the expert to list common symptoms or complaints that might require diagnosis. For each symptom, MOLE prompts for a list of possible explanations.
- MOLE then iteratively seeks out higher-level explanations until it comes up with a set of ultimate causes.
- During this process, MOLE builds an influence network similar to the belief networks.
- Whenever an event has multiple explanations, MOLE tries to determine the conditions under which one explanation is correct.
- The expert provides covering knowledge, i.e. the knowledge that a hypothesized event might be the cause of a certain symptom.
- MOLE then tries to infer anticipatory knowledge, which says that if the hypothesized event does occur, then the symptom will definitely appear. This knowledge allows the system to rule out certain hypotheses on the basis that specific symptoms are absent.

3.2 Refinement of the knowledge base

- MOLE now tries to identify the weaknesses of the knowledge base.
- One approach is to find holes and prompt the expert to fill them.
- It is difficult, in general, to know whether a knowledge base is complete, so instead MOLE lets the expert watch MOLE-p solving sample problems.
- Whenever MOLE-p makes an incorrect diagnosis, the expert adds new knowledge. There are several ways in which MOLE-p can reach the wrong conclusion.
- It may incorrectly reject a hypothesis because it does not feel that the hypothesis is needed to explain any symptom.
- It may advance a hypothesis because it is needed to explain some otherwise inexplicable hypothesis. Or it may lack differentiating knowledge for choosing between alternative hypotheses.
- For example, suppose we have a patient with symptoms A and B. Further suppose that symptom A could be caused by events X and Y, and that symptom B can be caused by Y and Z. MOLE-p might conclude Y, since it explains both A and B. If the expert indicates that this decision was incorrect, then MOLE will ask what evidence should be used to prefer X and/or Z over Y.

Module 2: Typical expert systems

4. MYCIN

- MYCIN, an early expert system, or artificial intelligence (AI) program, for treating blood infections.
- MYCIN is implemented in LISP. It was developed in the early 1970s at Stanford University and reached the stage of a research prototype.
- MYCIN worked with a knowledge base of about 600 rules and a simple inference engine.
- MYCIN assists physicians in the selection of appropriate antimicrobial therapy for hospital patients with bacteremia, meningitis, and cystitis infections.
- The system diagnoses the cause of the infection (e.g., the identity of the infecting organism is pseudomonas) using knowledge relating infecting organisms with patient history, symptoms, and laboratory test results.
- The system recommends drug treatment (type and dosage) according to procedures followed by physicians experienced in infectious disease therapy.
- MYCIN is a rule-based system employing a backward chaining control scheme. It includes mechanisms for performing certainty calculations and providing explanations of the system's reasoning process.

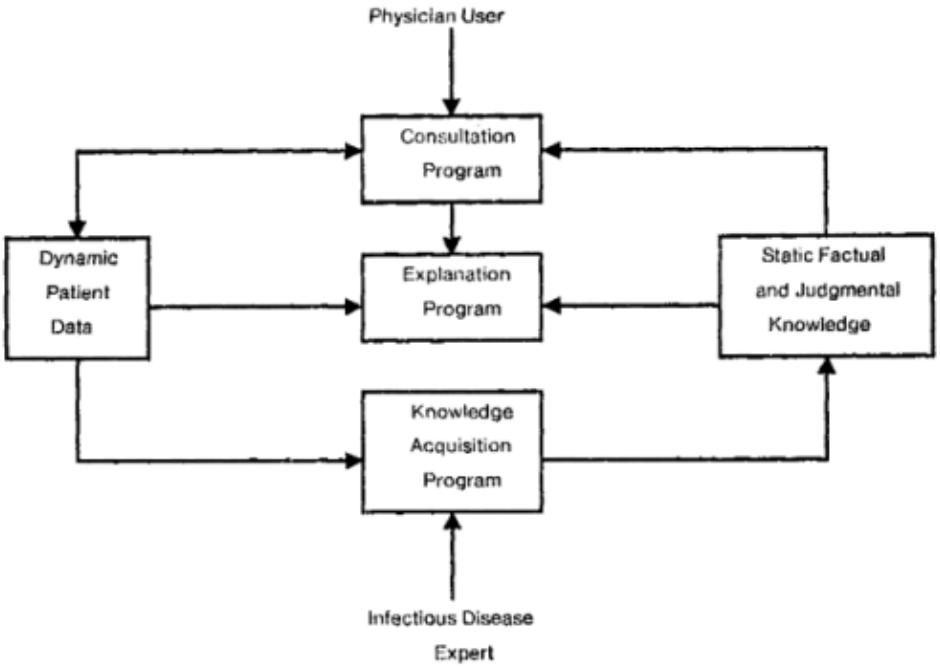


Figure 3: Structure of Mycin Program

- The MYCIN system comprises three major subprograms, as depicted in Figure 3 above.
  1. The Consultation Program
  2. Explanation Program
  3. Knowledge Acquisition Program
- The *Consultation Program* is the core of the system; it interacts with the physician to obtain information about the patient, generating diagnoses and therapy recommendations.
- The *Explanation Program* provides explanations and justifications for the program's actions.
- The *Knowledge-Acquisition* Program is used by experts to update the system's knowledge base.

5. DART

- DART assists in diagnosing faults in computer hardware systems using information about the design of the device being diagnosed.
- The system works directly from information about the intended structure and expected behavior of the device to help find design flaws in newly created devices.
- The system has been applied to simple computer circuits and the teleprocessing facility of the IBM 4331.
- DART uses a device-independent inference procedure that is similar to a type of resolution theorem proving, where the system attempts to generate a proof related to the cause of the device's malfunction.
- The system is implemented in MRS and was developed at Stanford University. It reached the stage of a research prototype.
- A typical, large-scale computer installation is composed of numerous subsystems including CPUs, primary and secondary storage, peripherals and supervisory software.
- Each of these subsystems, in turn, consists of a richly connected set of both hardware and software components such as disk drives, controllers, CPUs, memory modules, and access methods.
- Generally, each individual component has an associated set of diagnostic aids designed to test its own specific integrity.
- However, very few maintenance tools and established diagnostic strategies are aimed at identifying faults on the system or subsystem level.
- As a result, identification of single or multiple faults from systemic manifestations remains a difficult task.
- The non-specialist field service engineer is trained to use the existing component-specific tools as a result, is often unable to attack the failure at the systemic level.
- Expert assistance is then required, increasing both the time and cost required to determine and repair the fault.
- The design of DART reflects the expert's ability to take a systemic viewpoint on problems and to see that viewpoint to indicate a specific components, thus making more effective use of the existing maintenance capabilities.

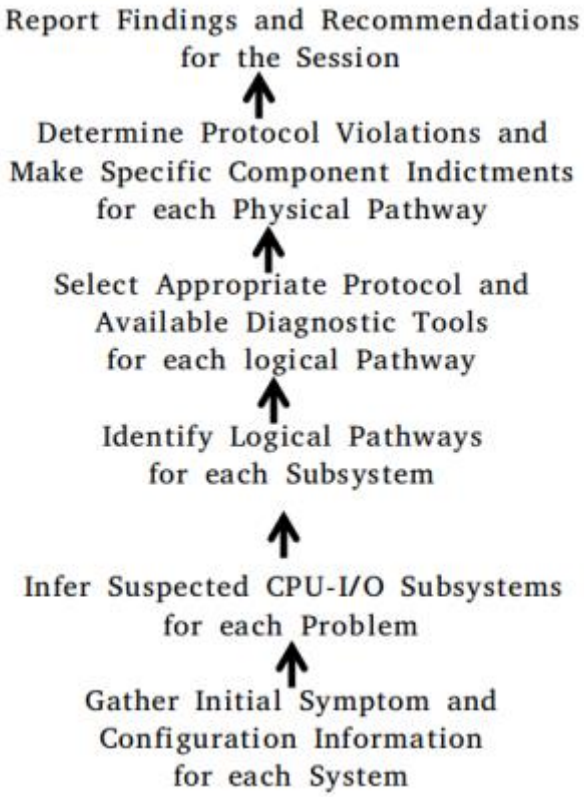


Figure 4: The DART diagnostic inference process

6. XCON

- XCON has the distinction of being one of the most mature and widely used expert systems currently operating on a commercial basis.
- It was developed jointly by an AI group at Carnegie-Mellon University (CMU) and an intelligent systems group at Digital Equipment Corporation.
- In 1980, XCON could configure VAX-11/780 computer systems; today it configures all VAX (Virtual Address eXtension) family systems for Digital Equipment Corporation (DEC) in its United States and European plants.
- XCON takes over the job previously performed by technical editors, people who examine a customer's purchase order and determine what computer components need to be substituted or added to make the order consistent and complete.
- XCON configures the systems at a very detailed level. For each order it determines necessary modifications, produces diagrams showing the spatial and logical relationships between the hundreds of components that comprise a complete system, defines cable lengths between system components, and handles other jobs usually relegated to skilled technicians.

- XCON performs at a level similar to that of an experienced technical editor, but it operates much faster. For example, although it takes the editor as long as 20 minutes to configure a system order, XCON can typically do it in less than a minute.
- The early versions of XCON were implemented in OPS4, later versions in OPS5 (rule-based or production system computer language).
- The DEC people call it XCON, while the CMU people prefer to call it R1. Both names refer to the same expert system.
- XCON was providing significant assistance to technical editors well before it reached its performance goal, which was set by DEC as configuring orders perfectly 90 to 95 percent of the time.

6.1 Functions of XCON System

- XCON is used to configure customer orders and to guide the assembly of these orders at the customer site. Using the customer order as input, it provides the following functionality:

1. Configures CPUs, memory, cabinets, power supplies, disk, and tapes and so on.
2. Determines and list cabling information.
3. List components ordered with configuration related comments.
4. Generate warning messages on issues affecting technical validity.

7. EXPERT SYSTEMS SHELLS

- Initially, each expert system that was built was created from scratch, usually in LISP. But, after several systems had been built this way, it became clear that these systems often had a lot in common.
- In particular, since the systems were constructed as a set of declarative representations (mostly rules) combined with an interpreter for those representations, it was possible to separate the interpreter from the domain-specific knowledge and thus to create a system that could be used to construct new expert systems by adding new knowledge corresponding to the new problem domain. The resulting interpreters are called shells.
- One influential example of such a shell is EMYCIN (for Empty MYCIN), which was derived from MYCIN.
- There are now several commercially available shells that serve as the basis for many of the expert systems currently being built. These shells provide much greater flexibility in representing knowledge and in reasoning with it than MYCIN did. They typically support rules, frames, truth maintenance systems, and a variety of other reasoning mechanisms.

- Early expert system shells provided mechanisms for knowledge representation, reasoning, and explanation. Later, tools for knowledge acquisition were added. But as experience with using these systems to solve real world problems grew, it became clear that expert system shells needed to do something else as well. They needed to make it easy to integrate expert systems with other kinds of programs.
- Expert systems cannot operate in a vacuum, any more than their human counterparts can. They need access to corporate databases, and access to them needs to be controlled just as it does for other systems. They are often embedded within larger application programs that use primarily conventional programming techniques.
- So one of the important features that a shell must provide is an easy-to-use interface between an expert system that is written with the shell and a larger, probably more conventional, programming environment.
- Components of Expert System shell
    1. Knowledge Base
    2. Knowledge Acquisition System
    3. Reasoning or Inference Engine
    4. User Interface
    5. Explanation Facilities
    6. Maintenance and Adaptation Tools
- Benefits of Expert System Shell in Artificial Intelligence
    1. Time and Cost Savings
    2. User-Friendly Interface
    3. Flexibility and Customization
    4. Built-in Features

QUESTION BANK

1 Mark: Objective Questions

Module 1: Architecture of expert systems

1. What is an expert system? [ ]

(a) A computer program that represents and reasons with knowledge of some specialist subject

(b) A computer program that emulates human emotions

(c) A computer program that mimics human physical movements

(d) A computer program that only provides advice to non-expert users

2. An expert system is a ___ which has the ability of decision-making like a human expert.

(a) Computer system (b) Operating system [ ]

(c) Transaction processing system (d) None of these

3. Which component is NOT part of an expert system? [ ]

(a) Inference Engine (b) Decision Maker

(c) Knowledge Base (d) User Interface

4. _____ are designed to solve complex problems by reasoning through bodies of knowledge.

(a) Computer systems (b) Expert systems [ ]

(c) Operating systems (d) Transaction processing systems

5. How does a non-expert user interact with an expert system? [ ]

(a) By only observing the expert system's behavior

(b) By providing physical demonstrations to the expert system

(c) By inputting complex algorithms into the expert system

(d) By asking questions or answering questions asked by the expert system

6. The _____ represents facts and rules. [ ]

(a) Knowledge base (b) Inference engine

(c) Operating system (d) None of these

7. The _____ applies the rules to the known facts to deduce new facts. [ ]

(a) Operating system (b) Knowledge base

(c) Inference engine (d) None of these

8. The expert systems are capable of _____. [ ]

(a) Decision making (b) Deriving a solution

(c) Interpreting input (d) All of these

9. In Expert System, User Interface is _____ [ ]

(a) Contains a computerized system

(b) Interacts between the user and the machine

(c) Friendly communication

(d) All of the mentioned above

10. In Expert System, Knowledge Acquisition means _____ [ ]

(a) How to get required domain knowledge by the expert system

(b) System maintenance

(c) System implementation

(d) None of the mentioned above

Explanation: In Expert System, Knowledge Acquisition means how to get required domain knowledge by the expert system.

11. Which university introduced Expert systems? [ ]

(a) Massachusetts Institute of Technology (b) University of Oxford

(c) Stanford University (d) University of Cambridge

Module 2: Typical expert systems

12. Which of the following expert system will use for diagnosis of blood infections? [ ]

(a) DART (b) XCON (c) MYCIN (d) DENDRAL

13. MYCIN system comprises of which of the following subprograms? [ ]

(i) The Consultation Program (ii) Explanation Program

(iii) Knowledge Acquisition Program (iv) Analysis Program

(a) (i) and (ii) only (b) (ii) and (iii) only

(c) (i), (ii) and (iii) only (d) (i), (ii), (iii) and (iv)

14. Which of the following expert system will use for diagnosing faults in computer hardware systems? [ ]

(a) DART (b) XCON (c) MYCIN (d) DENDRAL

15. Which of the following expert system will use for examine a customer's purchase order? [ ]

(a) DART (b) XCON (c) MYCIN (d) DENDRAL

16. A _____ is nothing but an expert system without knowledge base. [ ]

(a) Tools (b) Shell (c) Expert System (d) Knowledge

2 Marks: Short Answer Questions

Module 1: Architecture of expert systems

1. What is an expert system?
2. Draw architecture of expert systems.
3. List different members involved in development of an expert system.

Module 2: Typical expert systems

4. Specify different typical expert systems.
5. List various functions of XCON System.
6. What is shell?

5 Marks: Descriptive Questions

Module 1: Architecture of expert systems

1. Illustrate Architecture of expert systems.
2. What is expert system? Explain Roles of expert systems.
3. Describe Knowledge Acquisition Meta knowledge Heuristics.

Module 2: Typical expert systems

4. Explain Structure of MYCIN Program.
5. Describe how DART diagnostic inference processes.
6. Explain how XCON expert system will do job previously performed by technical editors to examine a customer's purchase order and determine what computer components need to be substituted.
7. What is shell? Explain expert systems shells.